

RAID

RAID, or Redundant Array of Independent Disks, is, quite simply, a setup that utilises more than one hard disk to either share or replicate data over hard disks. There are several different ways of implementing RAID and the type of RAID adopted dictates the relative merits and demerits. Motherboards today come with integrated SATA RAID controllers, and feature multiple versions of RAID.

We used one of our fastest drives, the 750 GB Seagate Barracuda, and for the RAID array, we used two Seagate Barracuda 7200.8 series 400 GB drives to show you the performance differences. Although RAID has many avatars, most home users need to know about only three of them.

RAID 0

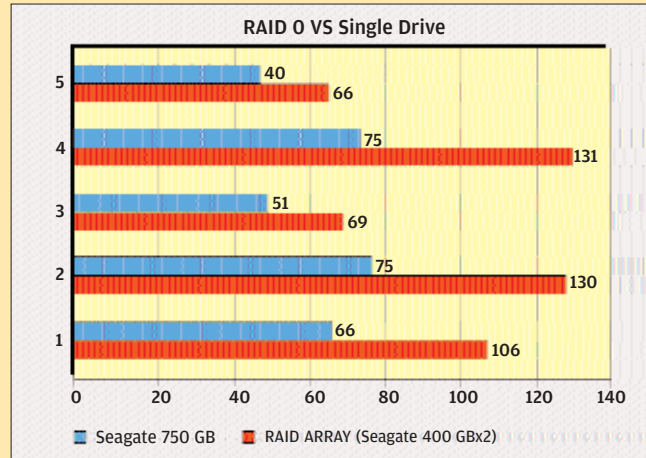
Also called Striping, this setup is for the speed fanatic. Striping doesn't have any sort of fault tolerance—a failure in any one of the striped drives will cause data loss. That said, drives of today are much less prone to crashes than their predecessors, and for pure speed, RAID 0 rules! Gamers and hardcore enthusiasts swear by their RAID 0 arrays.

RAID 0 needs at least two drives: data is broken down into blocks, and each block of data is written alternately to a separate hard drive. The load is, therefore, spread across a number of channels and drives. Another advantage of RAID 0 is that no parity calculation is involved, since there is no duplicating of data anywhere—so CPU overhead is minimised. RAID 0 is highly recommended for gamers, 2D and 3D professionals, and anyone who wants greater read/write performance from his storage subsystem.

RAID 1

Also called "Mirroring," RAID 1 offers data security via fault tolerance. It needs a minimum of two drives to implement: during a write operation in RAID 1, the same data is written to each of the drives simultaneously, so each drive contains a mirror of the data on the other drives. A read operation can be performed from any of these drives, so if one crashes, data can be accessed from the other.

The only disadvantage is that I/O performance is slower than with a single drive, and the parity computations—and therefore the CPU overheads—are much higher than with a single drive or even with striping. RAID 1 is very useful for users for whom data integrity means everything.



RAID 0+1

A combination of striping and mirroring, this features two or more clusters of RAID 0 arrays collectively configured as a mirrored array. RAID 0+1 requires a minimum of four hard drives to implement. Since this mode contains multiple striped segments, the I/O performance is blistering, while the presence of mirroring adds the much-needed parity to the mix. RAID 0+1 isn't without its demerits: it's very costly to implement, and besides, the failure of a single drive will in essence create a single, glorified RAID 0 array, and thereby defeat the purpose of parity completely.

Although implementing RAID seems like a costly affair, there are many users who may have a number of hard disks installed in their PCs, or even a couple of older hard disks lying around. With the advancements in today's onboard RAID controllers, it's no longer necessary to have drives of the same capacity, although "RAIDing" unequal-sized drives means you will lose the extra space the larger drive offers. Those with two or more similar-sized hard disks available are perfect candidates for experimenting with RAIDing. Remember, though: make a backup of all your essential data, as setting up a RAID array will destroy any data present on all the disks being used.

own for two reasons—first, it's much larger than the other drives, and second, it uses the Perpendicular Recording technology. So we are not giving it the winners title, even though the scores indicate otherwise.

Western Digital gets all the accolades in this category: the WD4000KS (400 GB) receives the *Digit Best Buy Gold*, and the WD5000KS (500 GB), the *Digit Best Buy Silver*.

Parallel ATA

In case it's got to be PATA for you, look no further than the Hitachi siblings, the Deskstar HDT722525DLAT80 and the Deskstar HDS725050KLAT80 (250 GB and 500 GB respectively). Bear in mind that the 500 GB Hitachi—our *Gold* award winner—maybe the best of the PATA drives, performance-wise, it is expensive at Rs 20,000. Go for it only if you simply must have the largest and fastest PATA drive out there. For just as much bang without a steep drop in your



Western Digital 500 GB
WD5000KS

bank balance, go with our *Silver* winner—the smaller Hitachi 250 GB, which, at Rs 5,700, offers you the best of both worlds.

Our Conclusion

We tested 26 hard drives this time round. Seven of these were the older PATA drives. With the majority of drives going SATA, it's the end of the road for PATA. All motherboards sold these days sport SATA interfaces, and with the possibility of CD/DVD drives going SATA, this seems fitting.

Much like how SATA debuted some years ago, Perpendicular Recording makes its way into the drive market, with the promise of larger density drives, equating to greater storage that you won't pay a premium for.

With Serial ATA promising as much as 600 MBps in the near future, the storage horizons seem promising indeed. As there is much good that has come before us, let there be more of the same! ■

michael_browne@thinkdigit.com